

DHARANI M

Senior AI Data Scientist | Gen AI Engineer | MLOps Specialist

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PROFESSIONAL SUMMARY

Data Scientist with 10+ years of hands-on experience building and deploying **production-grade AI/ML systems, Generative AI solutions, and MLOps pipelines** across **Banking, Healthcare, and Insurance** domains. Specialized in **LLM fine-tuning, RAG architectures, agentic AI frameworks (LangChain, AutoGen, CrewAI), and real-time ML systems on AWS, Azure, and GCP**. Proven track record of delivering **business-critical AI solutions** that improve fraud detection accuracy by **35%**, reduce operational costs by **40%**, and enable sub-second decision-making for **1M+ daily transactions**.

- Led end-to-end AI/ML lifecycle using **Python 3.12, SQL, scikit-learn, TensorFlow 2.x, and PyTorch 2.x**, delivering scalable models from ingestion to production monitoring.
- Built high-performance predictive models with **scikit-learn, PyTorch, and Fast.ai**, improving accuracy and driving measurable business outcomes.
- Designed distributed ML pipelines using **Apache Spark (PySpark)** integrated with **TensorFlow/PyTorch** for large-scale data processing and training.
- Developed and fine-tuned LLMs (**GPT-4/4o, LLaMA 2/3, Mistral**) using **OpenAI API, Hugging Face Transformers, and LangChain** for enterprise AI solutions.
- Engineered **RAG pipelines** using **FAISS, Pinecone, and ChromaDB**, improving contextual relevance and response accuracy.
- Designed **prompt engineering** and AI guardrail frameworks reducing hallucinations and enhancing model safety.
- Built LLM-powered applications including chatbots, document summarization, and intelligent search systems for business automation.
- Developed **NLP and knowledge graph** solutions using **spaCy, NLTK, Transformers, and Neo4j** enabling semantic search and Graph RAG capabilities.
- Delivered **computer vision** models (object detection, OCR, anomaly detection) using **OpenCV, YOLO, and** deep learning architectures.
- Built production-grade ML pipelines using **MLflow, Kubeflow, and Apache Airflow** with CI/CD automation and model versioning.
- Deployed scalable AI solutions on **AWS (SageMaker, S3, Lambda), Azure ML, and Google Vertex AI**, leveraging **Docker and Kubernetes** for orchestration.
- Designed ETL and real-time data pipelines using **Snowflake, SQL Server, and Apache Kafka**, and delivered insights via **Power BI, Tableau, and Plotly** while ensuring **Responsible AI** using **SHAP and LIME**.
- Collaborated with cross-functional teams (Data Engineering, DevOps, Product) in **Agile/Scrum** environments to deliver scalable, production-ready AI solutions.

TECHNICAL SKILLS

Programming & Scripting	Python, SQL (T-SQL, PostgreSQL, Snowflake SQL), PySpark, Scala, Java, NodeJS
Gen AI & LLMs	GPT-4o, Claude 3.5, LLaMA 3.1/3.2, Mistral 7B/8x7B, Gemini Pro, LangChain, LangGraph, AutoGen, CrewAI, Hugging Face Transformers, OpenAI API, Anthropic API, Ollama (local LLMs)
LLM Fine-Tuning & Training	LoRA, QLoRA, PEFT, RLHF, DPO, instruction tuning, Axolotl, Unsloth, vLLM, TensorRT-LLM
RAG & Vector DBs	Pinecone, Weaviate, Chroma, Qdrant, FAISS, LlamaIndex, embedding models (OpenAI, Cohere, Sentence-BERT), Graph RAG (Neo4j)
ML/DL Frameworks	PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, CatBoost, Fast.ai, Hugging Face (Transformers, Datasets, Accelerate)
Computer Vision	YOLOv8/v9, SAM (Segment Anything Model), YOLO-NAS, OpenCV, Detectron2, MMDetection, OCR (Tesseract, Azure Document Intelligence, AWS Textract)
NLP & Text Analytics	BERT, RoBERTa, T5, GPT-Neo, spaCy, NLTK, sentiment analysis, NER, topic modeling,

	semantic search
MLOps & Deployment	AWS SageMaker, Azure ML, Vertex AI, MLflow, Kubeflow, BentoML, Seldon Core, TorchServe, TensorFlow Serving, Docker, Kubernetes (AKS, EKS)
Cloud Platforms	AWS (SageMaker, Bedrock, Lambda, S3, Athena, Glue, EMR), Azure (ML Studio, OpenAI Service, Databricks, Synapse, Data Factory), GCP (Vertex AI, BigQuery, Dataflow)
Data Engineering	Databricks (Delta Lake, Unity Catalog), Snowflake, Apache Spark (PySpark), Kafka, Airflow, dbt, Azure Data Factory, AWS Glue, real-time streaming (Flink, Structured Streaming)
Databases	Snowflake, PostgreSQL, Azure SQL, MySQL, MongoDB, Neo4j (Graph DB), Redis, DynamoDB
Feature Engineering	Feature Store (Databricks, SageMaker Feature Store), automated feature engineering (Featuretools), drift detection
Model Monitoring & Explainability	SHAP, LIME, Integrated Gradients, Evidently AI, Arize AI, drift detection (data/concept drift), A/B testing frameworks
Data Visualization	Power BI, Tableau, Plotly, Matplotlib, Seaborn, Streamlit, Gradio
DevOps & CI/CD	GitHub Actions, Azure DevOps, Jenkins, GitLab CI/CD, DVC (Data Version Control), Terraform (IaC)
AI Security & Compliance	Prompt injection prevention, PII detection/redaction, Azure OpenAI content filters, HIPAA/SOC 2 compliance, model governance

PROFESSIONAL EXPERIENCE

Senior AI Data Scientist

First Republic Bank | New York, NY | **Aug 2023 – Present**

Real-Time Fraud Detection, AML Compliance & LLM-Powered Analytics

Leading the design and deployment of **production-grade Gen AI and ML systems** for fraud detection, AML compliance, and customer insights, processing **1M+ daily transactions** with **sub-second latency**. Architecting **LLM-powered agentic AI systems** for automated compliance workflows, document summarization, and intelligent customer service.

Key Achievements & Responsibilities:

- Architected and deployed **production RAG pipeline** using **LangChain + Pinecone + GPT-4** for **intelligent AML case summarization**, reducing analyst review time by **60%** and improving compliance audit accuracy
- Fine-tuned **LLaMA 3.1 (8B) using LoRA** on proprietary fraud transaction data, achieving **15% better F1 score** vs. GPT-3.5 for transaction classification while reducing inference costs by **70%**
- Built **agentic AI workflow orchestration** using **AutoGen + CrewAI** to automate KYC document verification, sanctions screening, and adverse media analysis, processing **10K+ cases/month**
- Developed **Graph RAG system with Neo4j** for entity relationship discovery in fraud networks, enabling detection of **previously unidentified money laundering patterns** involving 500+ entities
- Implemented **prompt engineering framework** with versioning, A/B testing, and hallucination detection guardrails, improving LLM response accuracy from **78% to 94%**
- Designed **Delta Live Tables + Spark Structured Streaming pipeline** for real-time fraud scoring, reducing detection latency from **5 minutes to <500ms** and preventing **\$12M in fraudulent transactions** annually
- Deployed **ensemble ML model (XGBoost + LightGBM + DNN)** for credit card fraud detection with **95% precision** and **40% reduction in false positives** vs. legacy rule-based system
- Built **feature store on Databricks** with **200+ engineered features** (velocity metrics, behavioral patterns, graph embeddings) supporting **real-time and batch inference** across 15+ ML models
- Implemented **online learning pipeline** for fraud models using **River (incremental learning)**, adapting to emerging fraud patterns within **24 hours** vs. **2 weeks** for traditional retraining
- Established **end-to-end MLOps pipeline** on **AWS SageMaker + MLflow**, enabling **CI/CD for 20+ production ML models** with automated retraining, A/B testing, and canary deployments
- Containerized **LLM inference services** using **vLLM + Kubernetes (EKS)**, achieving **3x throughput improvement** and **50% cost reduction** vs. SageMaker endpoints for high-volume use cases
- Implemented **model monitoring dashboard** using **Evidently AI + Grafana**, detecting data drift and triggering automated retraining workflows, maintaining model AUC **>0.92** in production
- Deployed **serverless AI agents** on **AWS Lambda + Bedrock** for ad-hoc compliance queries, reducing infrastructure costs by **65%** while maintaining **<2s response time**

- Developed **customer 360 analytics platform** combining **Snowflake + dbt + Power BI**, identifying **\$50M in cross-sell opportunities** through ML-driven propensity modeling
- Built **explainable AI framework** using **SHAP + LIME** for credit risk models, ensuring **regulatory compliance** and enabling **human-in-the-loop decision-making** for **high-risk loan applications**
- Created **automated compliance reporting pipeline** generating **regulatory reports (SAR, CTR)** using **LLM-powered document generation**, reducing manual effort by **80%**

Technologies: Python, PyTorch, TensorFlow, **GPT-4, LLaMA 3.1, Mistral 8x7B, LangChain, LangGraph, AutoGen, CrewAI, Pinecone, Weaviate, LoRA/QLoRA, vLLM, XGBoost, LightGBM, AWS (SageMaker, Bedrock, Lambda, S3, EKS), Databricks (Delta Live Tables, Feature Store, Unity Catalog), Kafka, Spark Streaming, Snowflake, MLflow, Kubeflow, SHAP, Evidently AI, Power BI, Tableau, Neo4j, Docker, Kubernetes**

AI Data Scientist – Insurance Analytics & Gen AI

Verisk Analytics (ISO) | Jersey City, NJ | May 2022 – July 2023

Claims Automation, Underwriting Intelligence & RAG-Powered Document Processing

Modernized insurance claims processing and underwriting workflows through **NLP, computer vision, and LLM-based automation**, processing **500K+ claims annually** and reducing manual review time by **50%**.

Key Achievements & Responsibilities:

- Developed **production RAG system** using **LangChain + Azure OpenAI (GPT-4) + Azure Cognitive Search** for **automated claims summarization**, processing **100K+ PDF documents/month** with **92% accuracy**
- Built **multi-modal AI pipeline** combining **Azure Document Intelligence (OCR) + BERT-based NER** for extracting structured data from unstructured claims documents (medical bills, police reports), achieving **88% field extraction accuracy**
- Fine-tuned **RoBERTa model** for insurance fraud detection from claim narratives, identifying **15% more suspicious claims** vs. keyword-based rules while reducing false alerts by **30%**
- Implemented **YOLOv8 + SAM (Segment Anything Model)** for **automated vehicle damage assessment** from photos, reducing adjuster review time by **40%** and improving damage estimate accuracy to **±5%**
- Deployed **anomaly detection model** using **Autoencoders + Isolation Forest** for identifying staged accident patterns in image metadata, uncovering **\$2M in fraudulent claims**
- Developed **gradient boosting ensemble (XGBoost + CatBoost + LightGBM)** for **churn prediction**, achieving **AUC 0.91** and enabling retention campaigns that reduced customer churn by **18%**
- Built **credit risk scoring model** using **logistic regression + SHAP explainability** for underwriting automation, approving **25% more low-risk applications** while maintaining **default rate <2%**
- Created **time-series forecasting pipeline** using **Prophet + LSTM** for claims volume prediction, improving resource allocation and reducing claim processing backlog by **35%**
- Architected **Snowflake data lakehouse** with **dbt transformations** and **Snowpark Python UDFs** for scalable feature engineering supporting **10+ ML models**
- Implemented **Delta Lake CDC pipelines** on Databricks for **real-time claims data ingestion** from legacy systems, ensuring **<5 min data freshness** for fraud detection dashboards
- Deployed models via **Azure ML managed endpoints** with **A/B testing framework**, enabling safe rollout of **3 major model updates** with **zero downtime**

Technologies: Python, PyTorch, TensorFlow, **GPT-4, RoBERTa, BERT, LangChain, Azure OpenAI Service, Azure Document Intelligence, YOLOv8, SAM, XGBoost, CatBoost, Snowflake, Snowpark, Databricks, Delta Lake, Azure ML, Azure Cognitive Search, Power BI, Tableau, dbt, Docker, Kubernetes**

Senior AI/ML Data Engineer – Healthcare Analytics

WellCare (Centene) | Tampa, FL | Feb 2021 – May 2022

HIPAA-Compliant AI/ML Pipelines, Clinical NLP & Predictive Healthcare Analytics

Designed and deployed **secure, scalable AI/ML infrastructure** for clinical decision support, patient risk stratification, and healthcare operations, processing **PHI for 5M+ members** while maintaining **HIPAA compliance**.

Key Achievements & Responsibilities:

- Built **clinical NLP pipeline** using **BioClinicalBERT + spaCy** for extracting medical entities (diagnoses, medications, procedures) from **unstructured clinical notes**, achieving **F1 score 0.89** on medical entity recognition
- Developed **FHIR-compliant data ingestion framework** on **Azure Synapse** for real-time processing of **HL7 messages** from 200+ hospitals, enabling **sub-5min clinical data availability**
- Implemented **GPT-4-powered discharge summary generation (PoC)** using **Azure OpenAI + RAG**, reducing physician documentation time by **30 min/patient** in pilot study

- Developed **hospital readmission risk model** using **survival analysis (CoxPH) + random forest**, identifying **high-risk patients with 82% recall**, enabling targeted care management for **50K+ patients**
- Built **medication adherence prediction model** using **time-series LSTM**, achieving **AUC 0.87** and reducing medication non-compliance by **22%** through proactive outreach
- Created **COVID-19 patient deterioration early warning system** using **XGBoost on real-time vitals data**, providing **6-hour advance warning** with **85% accuracy**
- Architected **HIPAA-compliant data lakehouse** on **Azure Data Lake + Databricks** with **Unity Catalog** for fine-grained access control and **audit logging** of all PHI access
- Implemented **PySpark-based ETL pipelines** processing **5TB+ daily healthcare data** from claims, pharmacy, labs, and clinical systems with **<2 hour SLA**
- Built **real-time patient monitoring dashboard** using **Kafka + Spark Streaming + Power BI**, enabling ICU staff to track **200+ vitals metrics** with **<30s latency**

Technologies: Python, PyTorch, BioClinicalBERT, spaCy (scispaCy), Azure OpenAI (GPT-4), Azure Synapse Analytics, Azure Data Lake, Databricks (Unity Catalog), PySpark, Kafka, FHIR/HL7 standards, XGBoost, LSTM, CoxPH (survival analysis), Power BI, Azure Key Vault, RBAC, Docker, Apache Airflow

Data Scientist – ML & Computer Vision

Mu Sigma | Bengaluru, India | **Jul 2018 – Sep 2020**

End-to-End ML/DL Solutions, Computer Vision & Advanced Analytics

Delivered **production-grade ML/DL models** across retail, manufacturing, and finance domains, including **anomaly detection, demand forecasting, and computer vision systems** achieving **10-35% performance improvements**.

Key Achievements:

- Developed **YOLOv5-based computer vision models** achieving **~96% precision**, reducing manual QA effort by **~70%**.
- Built **Transformer-based NLP models (BERT/GPT)** processing **1M+ reviews/month**, improving customer satisfaction by **~12%**.
- Implemented **time-series forecasting ensemble (Prophet, LSTM, XGBoost)** reducing MAPE by **~15%** and saving **~\$2.5M annually**.
- Designed real-time fraud detection pipelines using **GCP Dataflow + Pub/Sub**, handling **50K+ transactions/min** with **<200ms latency**.
- Developed and deployed **Generative AI models (GPT, LLaMA)** for automation, summarization, and anomaly detection.
- Built and optimized computer vision solutions using **PyTorch, TensorFlow, Hugging Face**, improving accuracy and reducing manual effort.
- Delivered end-to-end data science solutions including **data wrangling, EDA, feature engineering, model development, and deployment**.
- Developed scalable **ETL/ELT pipelines using Azure Data Factory and Databricks**, processing multi-terabyte datasets.
- Built real-time ingestion systems using **Pub/Sub and Dataflow** for low-latency analytics and streaming use cases.
- Applied advanced ML techniques including **time-series forecasting, clustering, SVM, survival analysis** for predictive insights.
- Improved model performance by **10–35%** using feature engineering, hyperparameter tuning, and cross-validation.
- Automated workflows and monitoring using **Azure Monitor, orchestration tools, and logging frameworks** to ensure reliability.
- Integrated ML models into applications via **REST APIs and cloud platforms** enabling real-time decision-making.

Technologies: Python, PyTorch, TensorFlow, YOLOv5, ResNet, BERT, Hugging Face Transformers, XGBoost, Prophet, **GCP (Vertex AI, BigQuery, Dataflow, Pub/Sub)**, Azure ML, Streamlit, Docker, Kubernetes

Data Scientist – Marketing & Supply Chain Analytics

Bridgei2i Analytics | Bengaluru, India | **Oct 2016 – Jun 2018**

Predictive Analytics, Customer Segmentation & Supply Chain Optimization

Built **ML models** for customer lifetime value prediction, churn prevention, and supply chain optimization, driving **\$5M+ in incremental revenue** through data-driven decision-making.

Key Achievements:

- Developed customer segmentation models using **K-Means, DBSCAN, and RFM analysis**, identifying high-value segments that increased campaign revenue by **~25%**.

- Built churn prediction models using **Logistic Regression and Random Forest**, achieving ~0.88 AUC and reducing churn by ~18%.
- Implemented inventory optimization using **Bayesian forecasting**, reducing stockouts by ~30% and overstock by ~22% across 500+ SKUs.
- Developed predictive ML solutions using **Python (pandas, NumPy, scikit-learn, NLTK)** for demand forecasting, segmentation, churn, and NLP insights.
- Applied advanced ML techniques including **Decision Trees, Random Forest, XGBoost, SVM, Bayesian models** for campaign optimization and supply chain risk analysis.
- Engineered scalable data pipelines using **BigQuery, Dataflow, Dataproc (Spark), HDFS, Azure Databricks** for large-scale data processing.
- Built real-time streaming architectures using **Apache Kafka and Apache Flink** enabling low-latency analytics for inventory and campaign tracking.
- Performed data preprocessing and feature engineering using **PySpark and Scala**, improving model accuracy and data quality.
- Evaluated models using **AUC-ROC, accuracy, and lift charts**, ensuring robust and actionable predictive performance.
- Translated business requirements into data-driven solutions including **NLP-based sentiment analysis and predictive modeling** for customer behavior.
- Designed executive dashboards using **Tableau and Power BI** to track marketing KPIs and supply chain performance.
- Integrated multi-source data using **SQL, Snowflake, Azure, and AWS**, enabling centralized analytics and strategic decision-making.

Technologies: Python, scikit-learn, XGBoost, **Kafka**, **Apache Flink**, **BigQuery**, **Dataproc (Spark)**, Tableau, Power BI, SQL

EDUCATION

Master of Science in Computer Science

University of Central Missouri | Jan 2021 – May 2022

Focus: Machine Learning, Deep Learning, Cloud Computing

Bachelor of Technology in Electronics and Communication Engineering

Acharya Nagarjuna University | 2013 – 2017

CERTIFICATIONS

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|---|------|
| • AWS Certified Solutions Architect – Professional | 2024 |
| • AWS Certified Machine Learning – Specialty | 2023 |